

Exercise (4.4.8 - Stefano Fochesatto). Let G be a group with subgroups H and K with $H \leq K$

- a. Prove that if H is characteristic in K and K is normal in G then H is normal in G .

Proof: Suppose that H is characteristic in K and K is normal in G . By definition we know that for all $g \in G$, $gKg^{-1} = K$. Said another way we know that for all $\phi \in \text{Inn}(G)$ we get $\phi(K) = K$. Note that since $\phi(K) = K$ we know that $\phi|_K \in \text{Aut}(K)$. Since H is characteristic in K , by definition for all $\sigma \in \text{Aut}(K)$ $\sigma(H) = H$. Thus $\phi|_K(H) = H$ and since $H \leq K$, $\phi(H) = H$ thus $H \trianglelefteq G$.

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- b. Prove that if H is characteristic in K and K is characteristic in G then H is characteristic in G . Use this to prove that the Klein 4-group V_4 is characteristic in S_4 .

Proof: Suppose $H \text{ Char } K$, and $K \text{ Char } G$. By definition for all $\sigma_g \in \text{Aut}(G)$ we know that $\sigma_g(K) = K$. Note that $\sigma_g|_K \in \text{Aut}(K)$. We also have by definition that for all $\sigma_k \in \text{Aut}(K)$, $\sigma_k(H) = H$. Therefore $\sigma_g|_K(H) = H$ and since $H \leq K$ we get that $\sigma_g(H) = H$ thus $H \text{ Char } G$.

We will show that V_4 is characteristic in S_4 by showing that $V_4 \text{ Char } A_4$ and $A_4 \text{ Char } S_4$. Note that since automorphisms preserve element order, for all $\sigma \in \text{Aut}(S_4)$, $\sigma(A_4)$ must map $\sigma(\langle (123), (124), (134), (234) \rangle) = \langle (123), (124), (134), (234) \rangle$ and the identity permutations must map together. This leaves only 3 elements of order 2 in A_4 left to map therefore it must be the case that $\sigma(\langle (12)(34), (13)(24) \rangle) = \langle (12)(34), (13)(24) \rangle$ (it can't be either the transpositions or the 4-cycles as we can only map 3 elements and those map in pairs). Thus for all $\sigma \in \text{Aut}(S_4)$, $\sigma(A_4) = A_4$. Note that $V_4 = \langle (12)(34), (13)(24) \rangle$, again since automorphisms preserve element order, for all $\sigma \in \text{Aut}(A_4)$, $\sigma(\langle (12)(34), (13)(24) \rangle) = \langle (12)(34), (13)(24) \rangle$ and identity permutations must map. Thus for all $\sigma \in \text{Aut}(A_n)$, $\sigma(V_4) = V_4$. Thus by the previous conclusion V_4 is characteristic in S_4 .

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- c. Give an example to show that if H is normal in K and K is characteristic in G then H need not be normal in G .

Solution: We have just shown that $V_4 \text{ Char } A_4$. Consider $H = \langle (12)(34) \rangle$ and note that $H \leq V_4$. Note that $H \trianglelefteq V_4$, since every $(ab)(cd)$ is it's own inverse we see that $(ab)(cd)(12)(34)(ab)(cd) = (12)(34)$. However H is not normal in S_4 , consider $(24)(12)(34)(24) = (14)(23)$. $\square$

Exercise (4.5.25 - Stefano Fochesatto). Prove that if G is a group of order 385 then $Z(G)$ contains a Sylow 7-subgroup of G and a Sylow 11-subgroup is normal in G .

Proof: Suppose G has order 385. Consider the prime factorization, $385 = (5)(7)(11)$. Therefore in G there exists a Sylow 5-subgroup order of order 5, a Sylow 7-subgroup P_7 , of order 7 and a Sylow 11-subgroup P_{11} , of order 11. From Sylow's Theorem we know that n_7 divides $(11)(5) = 55$, and it has form $n_7 = 1 + k(7)$. Note that $k > 7$ will not divide 55. Consider the following table which demonstrates that for $0 < k \leq 7$, n_7 does not divide 55.

$k(7)$	0	7	14	21	28	35	42	49
n_7	1	8	15	22	29	36	43	50
remainder	0	1	5	11	3	17	32	45

Thus $n_7 = 1$ and similarly to the previous problem $P_7 \trianglelefteq G$. Therefore $N_G(P_7) = G$ and recall that the quotient group $N_G(P_7)/C_G(P_7) \cong S_{P_7}$ for some $S_{P_7} \in \text{Aut}(P_7)$. Since P_7 is cyclic from Proposition 16 of Section 4.4 we know that $|\text{Aut}(P_7)| = \varphi(7) = 6$. Applying Lagrange's Theorem we know that $S_{P_7} \mid 6$ and therefore $|N_G(P_7)|/|C_G(P_7)| \mid 6$. Since P_7 is normal we get $385/|C_G(P_7)| \mid 6$, and since and since no divisor of 6 other than 1 divides 385 we get that $|N_G(P_7)|/|C_G(P_7)| = 1$ so therefore $|C_G(P_7)| = 385$ which implies that $C_G(P_7) = G$. Thus $P_7 \in Z(G)$.

From Sylow's theorem we know that n_{11} divides $(5)(7) = 35$, and it has the form $n_{11} = 1 + k(11)$. Clearly $12 \nmid 35$, $23 \nmid 35$ and $34 \nmid 35$, so $n_{11} = 1$ and therefore $P_{11} \trianglelefteq G$. $\square$

Exercise (4.5.44 - Stefano Fochesatto). Let p be the smallest prime dividing the order of the finite group G . If $P \in \text{Syl}_p(G)$ and P is cyclic prove that $N_G(P) = C_G(P)$.

Proof: Suppose that p is the smallest prime dividing $|G|$, with $P \in \text{Syl}_p(G)$ being cyclic. By definition there $|P| = p^\alpha$, and $|G| = p^\alpha m$. Note that since P is cyclic we have that $P \leq C_G(P) \leq N_G(P)$. Therefore the quotient group $|N_G(P)/C_G(P)|$ cannot divide p . Recall that the quotient group $N_G(P)/C_G(P) \cong S_P$ for some $S_P \in \text{Aut}(P)$. Since P is cyclic from Proposition 16 of Section 4.4 we know that $|\text{Aut}(P)| = \varphi(p^\alpha) = (p - 1)p^{\alpha-1}$. From Lagrange's Theorem we know that $|N_G(P)/C_G(P)| \mid p - 1$. Since $|N_G(P)/C_G(P)|$ cannot divide p we know that $|N_G(P)/C_G(P)| = p - 1$ or $|N_G(P)/C_G(P)| = 1$. If $|N_G(P)/C_G(P)| = p - 1$ since $N(G) \leq G$ it would imply that $p - 1 \mid |G|$ which is a contradiction since p is the smallest prime dividing $|G|$. Therefore $|N_G(P)/C_G(P)| = 1$ and thus $N(G) = C(G)$. $\square$